

HS 501: Global Health and Demography

Course Code	HS 501
Course Name	Global Health and Demography
L-T-P-C	3-0-2-4
Prerequisites	Elementary knowledge of probability and statistics; consent of the instructor
Intended for	Discipline Elective for MA Development Studies and Outside Discipline/Free Elective for Other Postgraduates and undergraduates (3rd and 4th year)
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Unit 1: Introduction to Global Health and Demography** (5 lecture Lectures + 4 lab Lectures)
 - Why should we study global health and demography?
 - Evolution of these subjects and their scope
 - Current population and health trends across the globe, South Asia, and India
 - Globalization, Population and health transformations
 - Sources of data for measuring population process and health
- **Laboratory Assignments**
 - Introduction to handling large scale health datasets such as the Demographic and Health Surveys; Gateway to Global Ageing data (with focus on LASI); National Sample Survey Organisation - Health consumption rounds; IHME- Global Burden of Disease data; Census of India; Sample Registration System Data
- **Demographic Measurement** (5 lecture Lectures + 12 lab Lectures)
 - Age-Sex structure- measurement and trends
 - Population aging and dependency ratios
 - Basic measures of fertility, mortality, nuptiality and migration
 - Estimating life expectancy using life tables
 - Stable population model
- **Laboratory Assignments**
 - Creating population pyramids
 - Direct indirect adjustment of Crude death rates for different age sex structures Adjusting health data for age-sex structure
 - Construction of abridged life table
 - Estimating under-five mortality rates using DHS data
- **Morbidity and Public Health** (10 lecture Lectures + 8 lab Lectures)
 - Measures of population health- prevalence and incidence
 - Measuring burden of disease-DALYs, DFLE, HLE
 - Basic concepts of epidemiology, study designs, risk of a disease,
 - Exploring the issues of association, causation, interaction, inference, confounding and generalisability
 - Survival analysis
- **Laboratory Assignments**
 - Calculation of disability rates
 - Calculation of disease free life expectancy
 - Calculation of prevalence rates for childhood diseases and undernutrition using DHS
 - Calculation of prevalence rates of major diseases using LASI, and the NSSO data
- **Demographic determinants of health** (7 lecture Lectures)
 - Health issues during infancy and childhood- nutrition, infectious diseases and breastfeeding etc
 - Adolescent health- sexual and reproductive health, anemia, BMI etc
 - Maternal health
 - Later adulthood and onset of non communicable diseases
 - Health of older adults- NCDs, disability, Pain and Cognition etc
 - Lifestyle and health behavior
 - Mental health across age groups

- Note: We will discuss gender differential in health issues at each age
- **Social Determinants of Health** (7 lecture Lectures + 4 lab Lectures)
 - Poverty, inequality and health
 - Mortality selection and poverty
 - Education and health- lifestyle choices, morbidity and health care utilization
 - Social capital and health- variations by income, education and ethnicity
 - Migration, urbanization and health;
 - Multiple deprivations and health
 - Social vulnerability and intergenerational impact of health
- **Laboratory Assignments**
 - Estimating the socio-economic gradient of any health issue (of choice) and quantifying their association
- **Space, Place and Health** (3 Lectures)
 - Space, culture and health-why is it essential to study health at disaggregated geographies
 - Impact of natural and indoor environment on health
 - Thinking about health at multiple level of geographies
- **Health Policy and Innovations** (5 Lectures)
 - Health systems across developed and developing nations- special focus on India and its states
 - National and state policies on child health, sexual and reproductive health, major diseases and family welfare
 - Universal health coverage- components and measurement
 - Health system strengthening and health care financing reforms
 - Policy and social innovations for modifying health behaviors, health care utilization and health care financing- with special focus on India

Text Books

1. Bhende, A., **Principles of Population Studies**, 7th Edition, Himalaya Publishing House, 1996.
2. Skolnik, R., **Global health 101**, Burlington, 2011.

Note: Classroom notes will be provided that will include list of research articles for reference

Reference books for laboratory sessions

1. Preston, S. H., Heuveline, P., & Guillot, M., **Demography: Measuring and modeling population processes**, Oxford: Blackwell Publishers Ltd., 2001.
2. Moultrie TA, RE Dorrington, AG Hill, K Hill, IM Timaeus and B Zaba (eds), **Tools for Demographic Estimation**, Paris: International Union for the Scientific Study of Population, 2013.
3. demographicestimation.iussp.org
4. Croft, Trevor N., Aileen M. J. Marshall, Courtney K. Allen, et al., **Guide to DHS Statistics**, Rockville, 2018.

References

1. Rothman, K. J., Greenland, S., & Lash, T. L., **Modern epidemiology** (Vol. 3), Wolters Kluwer Health/Lippincott Williams & Wilkins, 2008.
2. Jacob S. Siegel and David a. Swanson, **The Methods and Materials of Demography**, 2nd Edition, Elsevier Science, USA, 2004.
3. Lee E T, **Statistical Methods for Survival Data Analysis**, 2nd Edition, John Wiley & Sons.
4. Ladusingh, L, **Survey Sampling Methods**, PHI Learning, 2018.
5. Hsiao, W.C., **What is a health system? Why should we care?**, Harvard School of Public Health Working Paper, 2003.
6. Mosley, W. H. and L. C. Chen, Analytical framework for the study of child survival in developing countries, **Population and Development Review** 10 (Supplementary Copy), 1984.
7. Murray, C. J. L., **Quantifying the Burden of Disease: The Technical Basis for Disability Adjusted Life Years**, Bulletin of the WHO, Vol. 72(3), pp.429-445, 1994.
8. Berkman, L. F., Kawachi, I., & Glymour, M. M. (Eds.), **Social epidemiology**, Oxford University Press, 2014.
9. Deaton, A., **The great escape**, Princeton University Press, 2013.
10. Morland, P., **The human tide: how population shaped the modern world**, Hachette UK, 2019.